

# Electrodynamics

## Independence Semester 2024

### Homework 5

November 14, 2024

#### Problem 1

a) We know that Legendre polynomials,  $P_\ell(x)$ , satisfy the differential equation

$$(1 - x^2) \frac{d^2 P_\ell(x)}{dx^2} - 2x \frac{dP_\ell(x)}{dx} + \ell(\ell + 1) P_\ell(x) = 0. \quad (1)$$

Using this information prove that  $\int_{-1}^1 dx P_\ell(x) P_n(x) = 0$ .

b) The generating function of Legendre polynomials is given by

$$G(x, t) = (1 - 2xt + t^2)^{-1/2} = \sum_{\ell \geq 0} t^\ell P_\ell(x). \quad (2)$$

Using the generating function and the results of part a) prove that  $\int_{-1}^1 dx [P_\ell(x)]^2 = \frac{2}{2\ell + 1}$ .

#### Problem 2

Express  $f(x) = x^4 + 3x^3 + x^2 + 5x - 2$  in terms of Legendre polynomials.

#### Problem 3

A thin nonconducting rod of length  $L$  carries a uniformly distributed charge  $Q$  and is oriented along the  $z$ -axis (see Figure 1).

a) Find the electrostatic potential  $\phi$  due to the charged rod for any point on the  $z$ -axis with  $z > L/2$ .

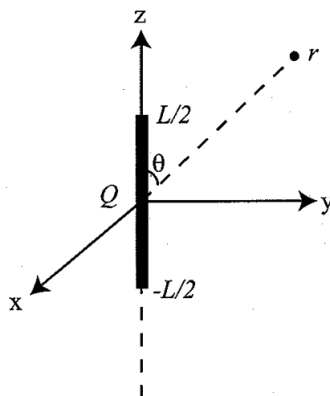


Figure 1: Figure for problem 3

- b) Find  $\phi(r, \theta)$  for all  $|r| > L/2$ , where  $r$  and  $\theta$  are the usual spherical coordinates. Note that since this problem has azimuthal symmetry,  $\phi$  is independent of  $\varphi$ .
- c) Check your result in b) in the limit  $r \gg L$ .

## Problem 4

A spherical shell of radius  $R$  is centred at the origin of our coordinate system. The electric potential on the shell is known to be

$$\phi(\theta, \varphi) = V_0 \sin \theta \cos \phi, \quad (3)$$

where  $V_0$  is a constant, and we use the usual spherical polar coordinates.

- a) Find  $\phi(r, \theta, \varphi)$  anywhere, both inside and outside the sphere. Assume that the zero of  $\phi$  is fixed by requiring  $\phi$  to approach zero at spatial infinity.

[**Hint:** This problem can be solved by using standard spherical harmonics]

- b) Suppose instead that the potential on the shell is given by

$$\phi(\theta, \varphi) = V_0 \sin^2 \theta \sin^2 \phi, \quad (4)$$

Again find  $\phi(r, \theta, \varphi)$  everywhere, both inside and outside the sphere.

## Problem 5

Let us consider a sphere of radius  $a$ , where one hemisphere of the boundary surface was held at a constant potential  $+V$ , while the other hemisphere was held at potential  $-V$ ,

i.e.,

$$\begin{aligned}\phi(a, \theta, \varphi) &= +V, & 0 \leq \theta < \pi/2 \\ &= -V, & \pi/2 \leq \theta \leq \pi.\end{aligned}\tag{5}$$

Let us assume that you evaluated the potential on the positive  $z$ -axis for  $z > a$  to be

$$\phi(z) = V \left( 1 - \frac{z^2 - a^2}{z\sqrt{z^2 + a^2}} \right).\tag{6}$$

- a) Now, using the symmetry of the problem, evaluate  $\phi(z)$  on the negative  $z$ -axis for  $z > a$ . You can also evaluate this result explicitly.
- b) Write down the results for  $\phi(z)$  for both the positive and negative  $z$ -axis in a single compact expression.
- c) Using the expressions for  $\phi(z)$  for  $\phi(z, 0)$  and  $\phi(|z|, \pi)$  in equations 4.60D and 4.60E in the notes and the results of part a) and b), evaluate the coefficients  $B_\ell$  and  $\tilde{B}_\ell$ .
- d) Evaluate  $\phi(r, \theta)$  using the off-axis extrapolation method, valid in the region  $r > a$ .
- e) Check that the general off-axis potential indeed changes sign under reflection in  $z = 0$  plane, i.e., under  $\theta \rightarrow \pi - \theta$ . Explain your calculation, particularly emphasizing on what property of  $P_\ell(x)$  allows us to establish the result.
- f) Repeat the exercise of parts a) to d) to find the potential  $\phi(r, \theta)$  valid in the region  $r < a$ , i.e., in the interior region of the spherical boundary at  $r = a$ .
- g) Show that the results of parts d) and f) are indeed related by the method of inversion by keeping terms in both power series upto the order  $r^7$ .